

Data Visualisation with Matplotlib & Seaborn

A chart that needs explaining has already failed. By the end of this lesson, every plot you ship will land in five seconds — title, axes, units, and the one insight a human came for.

L7

WHAT'S INSIDE

- 01** Matplotlib's anatomy — figure, axes, artists
- 02** Four core chart types — line, bar, scatter, hist
- 03** Seaborn — statistical plots, less code
- 04** Customisation — titles, labels, colours
- 05** Saving publication-ready figures

WHY THIS LESSON MATTERS

Numbers in a table convince no one. Charts decide meetings.

WHY

Real

The chart is the deliverable — not the notebook.

SCALE

Daily

Analysts ship 5–10 plots a day across standups and decks.

SPEED

Now

Matplotlib + Seaborn cover 90% of what you'll plot for years.

IMPACT

Yours

A clear chart on slide 3 wins the room before slide 4 loads.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Explain Matplotlib's figure-axes mental model in one sentence.

2

Build line, bar, scatter, and histogram plots from any DataFrame.

3

Pick the right Seaborn plot for a statistical question.

4

Add titles, axis labels, and legends so a stranger understands the chart.

5

Save figures at the right size and DPI for slides, reports, and web.

AGENDA

What we'll cover today

Four short parts. Each one builds on the last.

01

The Matplotlib mental model

Figure → Axes → plot. The only model you need.

02

Four core charts

Line, bar, scatter, histogram — when to use which.

03

Seaborn for stats

boxplot, violin, heatmap, pairplot.

04

Polish & save

Titles, labels, colours, DPI.

01

PART 1 OF 4

The Matplotlib mental model

Every plot is a Figure that holds Axes. The Axes is where the chart actually lives. Master those two words and the rest is recipe.

CONCEPT 1 OF 4

Figure, Axes, and the recipe behind every plot

```
import matplotlib.pyplot as plt
```

```
fig, ax = plt.subplots(figsize=(8, 5))  
ax.plot(df['month'], df['orders'])  
ax.set_title('Monthly orders, FY24')  
ax.set_xlabel('Month'); ax.set_ylabel('Orders')  
plt.show()
```

Figure = the canvas. Axes = the chart inside it.

DEFINITION

Figure & Axes

Figure is the page. Axes is the chart drawn on the page.
Almost every Matplotlib call is a method on an Axes object.

EXAMPLE

Always start with `fig, ax = plt.subplots()`. Never use the global `pyplot` state in production code.

CONCEPT 2 OF 4

Four core chart types — pick by question, not data type

`ax.plot(x, y)` → trend over time

`ax.bar(cats, vals)` → compare across categories

`ax.scatter(x, y)` → relationship between two variables

`ax.hist(values, bins=30)` → distribution of one variable

The question decides the chart. The data type just enables it.

KEY POINTS

- Line — change over time
- Bar — compare categories
- Scatter — two-variable relationship
- Histogram — distribution shape

CONCEPT 3 OF 4

Seaborn — statistical plots in one line

```
import seaborn as sns
```

```
sns.boxplot(data=df, x='city', y='price')
```

```
sns.histplot(df, x='price', hue='category', kde=True)
```

```
sns.heatmap(df.corr(numeric_only=True), annot=True)
```

```
sns.pairplot(df, hue='segment')
```

Pandas DataFrame in. Statistical chart out. No loops needed.

DEFINITION

Seaborn

A high-level wrapper over Matplotlib. Built for statistical charts from tidy DataFrames — categorical splits, distributions, correlations.

EXAMPLE

Seaborn for stats; Matplotlib for fine control. They compose — every Seaborn call returns an Axes you can keep editing.

CONCEPT 4 OF 4

Polish — a chart a stranger can read in 5 seconds

```
ax.set_title('Avg basket size by city, Q1 FY25')  
ax.set_xlabel('City'); ax.set_ylabel('₹ per order')  
ax.legend(title='Segment', loc='upper right')  
sns.set_theme(style='whitegrid', palette='deep')
```

```
fig.savefig('out.png', dpi=150, bbox_inches='tight')
```

Title answers 'what?'. Axes answer 'in what units?'.

KEY POINTS

- Title = the one-sentence takeaway
- Axis labels must include units (₹, %, count)
- Limit to 5 colours; never red/green together
- Save at 150 DPI, tight bbox, PNG for slides

02

PART 2 OF 4

Engage and reflect

Pause. Pick up the dataset you cleaned in Lesson 4 and turn one question into one chart that a non-analyst would understand.

✦ THINK • PAIR • SHARE

ACTIVITY

Three plots from one DataFrame

PROMPT

Open the cleaned e-commerce DataFrame from Lesson 4. Pick three questions a CFO would ask, then build the smallest chart that answers each. One Matplotlib, one Seaborn, one customised. Five minutes. No defaults — every chart must have a title and labelled axes.

HOW TO RUN IT

- 1 Think (1 min) — write three CFO questions.
- 2 Plot (3 min) — one chart per question.
- 3 Pair (1 min) — can your partner read it cold?
- 4 Share — one team posts their best plot.

REFLECT & APPLY

Three questions before you close this lesson

60 seconds each. Answer honestly — there's no scoreboard here.

Q1

Which of your last three charts at work could a stranger read in five seconds?

Q2

When did a chart default — bin width, colour, axis range — mislead a decision?

Q3

What's the one chart you'll redraw tomorrow with the labels you skipped last time?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. No googling.

Which object holds the actual chart in Matplotlib?

A

Figure

B

Axes

C

Canvas

D

Plot

Which object holds the actual chart in Matplotlib?



CORRECT ANSWER

Axes

WHY

Figure is the page; Axes is the chart drawn on it. Almost every chart-editing method (`set_title`, `set_xlabel`, `plot`) belongs to an Axes.

You want to see the distribution of order values. The right chart is:

A

Line plot

B

Bar chart

C

Histogram

D

Pie chart

You want to see the distribution of order values. The right chart is:



CORRECT ANSWER

Histogram

WHY

Histograms bucket a continuous variable into bins and show the shape of the distribution — skew, peaks, outliers — in one glance.

Quickest way to compare a numeric variable across categories with median + outliers:

A

`sns.boxplot`

B

`sns.heatmap`

C

`sns.scatterplot`

D

`ax.bar`

Quickest way to compare a numeric variable across categories with median + outliers:



CORRECT ANSWER

sns.boxplot

WHY

boxplot shows the median, IQR, whiskers, and outliers per category in one Seaborn call — exactly the comparison a category-vs-numeric question needs.

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week. The chart that goes on slide 3 of your next deck.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Figure is the page; Axes is the chart — that's the whole mental model.
- 2 Pick the chart by the question — line, bar, scatter, histogram each answer a different one.
- 3 Seaborn for statistical patterns; Matplotlib for fine control. They compose.
- 4 Title, axis labels, units — never optional. The chart must read in five seconds.
- 5 Save at 150 DPI with `bbox_inches='tight'` for slide-ready figures.

WORKED EXAMPLE

An Indian e-commerce dashboard in four charts

THE SCENARIO

```
fig, axes = plt.subplots(2, 2, figsize=(10, 7))  
axes[0,0].plot(df['month'], df['orders']) # trend  
axes[0,1].bar(top5.index, top5.values)   # top categories  
sns.histplot(df['price'], bins=40, ax=axes[1,0]) # distribution  
sns.scatterplot(data=df, x='price', y='rating', ax=axes[1,1])
```

THE TAKEAWAY

"Four charts on one page — and a CFO reads them in under a minute."

WHAT TO NOTICE

1

Line — orders climb after Diwali

2

Bar — fashion + electronics own 60%

3

Hist — price skews to ₹500–₹1500

4

Scatter — no link between price & rating

Mini assignment — a four-chart e-commerce snapshot

Take the cleaned DataFrame from Lesson 4. Build a 2×2 figure: trend, top categories, price distribution, and price-vs-rating scatter.

DELIVERABLES

- Use `plt.subplots(2, 2, figsize=(10, 7))`.
- Title + labelled axes + units on every panel.
- One Seaborn chart, three Matplotlib charts.
- Save as PNG at `dpi=150, bbox_inches='tight'`.

WHAT 'GOOD' LOOKS LIKE

One PNG. A non-analyst on your team can answer "how is the business doing?" in 60 seconds — without you in the room.

ONE THING TO REMEMBER

“

A chart that needs explaining
has already failed.

— AI Learning Hub

END OF LESSON 7

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Build a chart with `fig, ax = plt.subplots()` — no global state.
- 2 Pick line / bar / scatter / hist by the question, not the data type.
- 3 Drop a Seaborn one-liner when the question is statistical.
- 4 Ship a chart a stranger reads in five seconds.
- 5 Save figures at 150 DPI for slides and reports.

NEXT → Capstone Project — Indian E-commerce Wrangling. All four formats, one analysis, one dashboard. 90 min.